



Air Quality Construction Permit

Permit Number: 13-A-187-S3

Plant Number: 75-05-005

Company: Plymouth Energy, LLC

Contact Person:

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EHS Manager

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22056 K-42

Merrill, IA 51038

Responsible Party:

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Plant Manager

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22234 K-42

Merrill, IA 51038

Permitted Equipment

Emission Point ID: SV16

Table 1 – Emission Unit(s) and Control Equipment:

EU ID	Description	Maximum Rated Capacity	Control Equipment Description and ID
EU16	Batch Mash Fermenter No. 1	762,550 gallons, 112.5 gallons mash/min	CO ₂ Scrubber (CE C07), and Polishing Scrubber (CE C14)
EU17	Batch Mash Fermenter No. 2	762,550 gallons, 112.5 gallons mash/min	
EU18	Batch Mash Fermenter No. 3	762,550 gallons, 112.5 gallons mash/min	
EU19	Beer Well	990,007 gallons	
EU42	Batch Mash Fermenter No. 4	762,550 gallons, 112.5 gallons mash/min	

Equipment Location: 22056 K-42

Merrill, IA 51038

Issuance of this permit shall not relieve the owner or operator of the responsibility to comply fully with applicable provisions of the State Implementation Plan (SIP), and any other requirements of local, state, and federal law.

Table 2 – Project Issuance Information

Project Number	Project Description	Stack Testing	Issuance Date
23-293	Remove pre-fermentation/distillation equipment from equipment list	Yes	2/1/2024

Under the Direction of the Director of the
Department of Natural Resources

PERMIT CONDITIONS

1. Emission Limits

The owner or operator is required to report all emissions as required by law, regardless of whether a specific emission limit has been established in this permit. The following emission limits shall not be exceeded:

Table 3 – Emission Limits

Pollutant	lb/hr ¹	tons/yr ²	Other Limits	Reference/Basis
Particulate Matter (PM) – State	0.20 ³	NA	0.1 gr/dscf ⁴	567 IAC 23.4(7), Limit PTE
PM ₁₀	0.20 ⁴	NA	NA	NAAQS
PM _{2.5}	NA	NA	NA	NA
Opacity	NA	NA	40% ^{5, 6}	567 IAC 23.3(2)"d"
Volatile Organic Compounds (VOC)	11.21 ³	NA	NA	Title V Synthetic Minor
(Single HAP)	1.32 ⁷	NA	NA	NESHAP Area Source
(Total HAP)	2.07 ⁷	NA	NA	NESHAP Area Source

¹ The emission limit is expressed as the average of three runs.

² The emission limit is based on a 12-month rolling total.

³ The emission limits were established in Project Numbers 23-293 and 13-041 to reduce potential emissions of Plymouth Energy below the "major source" threshold for the purposes of the Prevention of Significant Deterioration (PSD) and Title V programs.

⁴ The emission rate was used in the dispersion modeling analysis conducted for Project 09-454. The results from the analysis show that this emission rate will cause predicted concentrations that are less than the National Ambient Air Quality Standards (NAAQS) for this pollutant.

⁵ The emission limit is based on a six minute average.

⁶ An exceedance of the indicator opacity of "*No Visible Emissions (NVE)*" will require the owner or operator to promptly investigate the emission unit and make corrections to operations or equipment associated with the exceedance. If exceedances continue after the corrections, the Department may require additional proof to demonstrate compliance (e.g., stack testing).

⁷ HAP emission limits established in Project Number 23-293 to reduce potential emissions of Plymouth Energy, LLC below "major source" threshold for the purposes of Title V Operating Program and NESHAP applicability as defined in 40 CFR Part 63 §63.2.

2. Compliance Demonstration(s)

If an initial stack test is specified in the "Compliance Demonstration" table, the owner or the owner's authorized agent shall demonstrate compliance with the emission limitations contained in Condition 1 (Emission Limits) within the applicable time period specified below:

- Within 60 days after achieving the maximum production rate but not later than 180 days after the initial startup date of the proposed equipment for the addition of new equipment or the physical modification of existing equipment or control equipment.
- Within 90 days of the issuance of this permit if there is no physical modification to any emission units or control equipment.

If any additional stack testing beyond an initial test (i.e. quarterly, semi-annual, annual, etc.) is required in "Compliance Demonstration" table, the owner or the owner's authorized agent shall demonstrate compliance with the emission limitations contained in Condition 1 (Emission Limits) as specified in the "Compliance Demonstration Table." See Conditions 12.A.(4) and 12.B.(5) for notification and reporting requirements.

If stack testing is required, the owner or the owner's authorized agent shall use the test method and run time listed in the "Compliance Demonstration" table unless another testing methodology is approved by the Department before testing.

Table 4 - Compliance Demonstrations

Pollutant	Compliance Methodology	Frequency	Test Run Time	Test Method
PM – State	None	NA	1 hour	40 CFR 60, Appendix A, Method 5 40 CFR 51, Appendix M, Method 202
PM ₁₀	None	NA	1 hour	40 CFR 51, Appendix M, 201A with 202
PM _{2.5}	None	NA	1 hour	40 CFR 51, Appendix M, 201A with 202
Opacity	None	NA	1 hour	40 CFR 60, Appendix A, Method 9
VOC	Stack Test ^{1, 2}	Annually ³	1 hour	40 CFR 63, Appendix A, Method 320 or 40 CFR 60, Appendix A, Method 18
HAP	Stack Test ^{2, 4}	Annually ³	1 hour	40 CFR 63, Appendix A, Method 320 or 40 CFR 60, Appendix A, Method 18

¹ VOC compliance testing may be determined using the sum of the Method 320 or Method 18 results.

² Testing of this stack shall be conducted in a manner to verify compliance with all emission limits with all equipment operating under maximum normal conditions. Testing shall be completed during the months of June, July, or August.

³ Performance testing shall be conducted once every twelve months with a minimum of six months between tests. For the Single HAP limit, this requirement shall apply to each individual HAP tested.

⁴ Acrolein, acetaldehyde, formaldehyde, and methanol shall be tested for specifically. HAP compounds that test below the detection limits shall be assumed to be emitted at a rate equal to the detection limit.

Each emissions compliance test must be approved by the Department. Unless otherwise specified by the Department, each compliance test for an air pollutant, excluding opacity, shall consist of three separate runs. The arithmetic mean of three acceptable test runs shall apply for compliance, unless otherwise indicated by the Department.

Opacity compliance tests shall consist of a minimum of three, 1-hour runs of observations. Opacity shall be determined as the average of any 24 consecutive, 15-second observations from the data set. The opacity observation duration and averaging time requirements apply unless otherwise specified by federal rule, specified in this permit, or granted prior written approval by the Department.

In accordance with 567 IAC 25.1(7)"a":

- (1) At the Department's request, a pretest meeting shall be held not later than 15 days before the owner or operator conducts the compliance demonstration. A testing protocol shall be submitted to the Department no later than 15 days before the owner or operator conducts the compliance demonstration. Representatives from the Department shall attend this meeting, along with the owner and the testing firm, if any. It shall be the responsibility of the owner to coordinate and schedule the pretest meeting.
- (2) A representative of the Department shall be permitted to witness the stack tests. In order to allow a Department representative the opportunity to observe a stack test, each test must begin on a weekday, between the hours of 6 am to 6 pm. Alternative stack test times may be granted through written Department approval prior to testing.

- (3) The Department shall reserve the right to impose additional, different, or more detailed testing requirements.

The owner shall be responsible for the installation and maintenance of test ports.

The unit(s) being sampled shall be operated in a normal manner (i.e. not under startup or shutdown conditions) at

- (a) its maximum continuous production or operating rating as rated by the equipment manufacturer, which is listed on either the first page or Condition 3, Emission Point Characteristics, of this permit, or
- (b) a permitted rating listed elsewhere in this permit that is less than the maximum continuous production or operating rating as rated by the equipment manufacturer.

If the compliance test is conducted at less than (a) or (b) above then the owner or operator shall either retest the unit(s) under the conditions of (a) or (b) above or the Department may require additional information or action to determine the unit(s) compliance status with applicable emission limits. This information or action includes, but is not limited to, a permit amendment, additional testing, continuous monitoring, and operating data.

3. Emission Point Characteristics

This emission point shall conform to the specifications listed in the table below.

Table 5 – Emission Point Parameters

Parameter	Value
Stack Height (feet from the ground)	75
Discharge Style	Vertical Unobstructed
Stack Outlet Dimensions (inches)	14
Exhaust Temperature (°F)	Ambient
Exhaust Flowrate (scfm)	6,000

The temperature and flowrate are intended to be representative and characteristic of the design of the permitted emission point. The Department recognizes that the temperature and flow rate may vary with changes in the process and ambient conditions. If it is determined that any of the emission point characteristics above are different than the values stated, the owner or operator shall submit a request either by electronic mail or written correspondence to the Department within 30 days of the discovery to determine if a permit amendment is required, or submit a permit application requesting to amend the permit.

4. Federal Standards

A. New Source Performance Standards (NSPS):

These emission units are not subject to any NSPS subparts at this time as there are no applicable subparts for their source category.

NOTE: The absence of the inclusion of any NSPS requirements as part of this permit does not relieve the owner or operator from any obligation to comply with all applicable NSPS conditions.

B. National Emission Standards for Hazardous Air Pollutants (NESHAP):

This facility is of the source category for Subpart VVVVVV (*Chemical Manufacturing Area Sources*; 40 CFR §63.11494 – §63.11503). However, this emission unit is not subject as it has been determined that dry mill ethanol plants do not have a listed Table 1 HAP in greater than 0.1% concentration.

NOTE: The absence of the inclusion of any NESHAP requirements as part of this permit does not relieve the owner or operator from any obligation to comply with all applicable NESHAP conditions.

5. Operating Requirements with Associated Monitoring and Recordkeeping

Unless specified by any federal regulation, all records as required by this permit shall be available on-site for a minimum of two years and shall be available for inspection by the Department. Records shall be legible and maintained in an orderly manner. The operating requirements and associated recordkeeping for this permit shall be:

- A. The CO₂ Scrubber (CE C07) and Polishing Scrubber (CE C14) shall each have a minimum scrubber liquid flow rate that is equal to or greater than the average liquid flow rate, at the inlet to the wet scrubber, as measured during the most recent performance test that demonstrated compliance with all applicable emission limits. The average liquid flow rate shall be based on a 3-hour block averaging period. This requirement shall not apply on the days when the equipment the scrubber's control and the scrubbers are not in operation.
 - (1) The owner or operator shall record the scrubber liquid (water) flow rate on a continuous basis.
 - (2) The owner or operator shall calculate and record the liquid flow rate based on a 3-hour block average.
 - (3) If the flow rate deviates below the minimum flow rate required (i.e., the average liquid flow rate observed during the applicable performance test), then the facility shall record the time, date and actions taken to correct the situation and when the flow rate is back above the minimum flow rate required.
- B. Any additive added to a scrubber liquid (CO₂ Scrubber (CE C07) or Polishing Scrubber (CE C14)) during a compliance test to enhance the efficiency of the scrubber shall be added at a rate greater than or equal to the rate recorded during the applicable operating performance test that demonstrated compliance with all applicable emission limitations. The additive feed rate shall be based on a 3-hour block averaging period. This requirement shall not apply on the days when the equipment the scrubber's control and the scrubbers are not in operation.
 - (1) The owner or operator shall record the rate of additive added (additive feed rate) to the scrubber liquid on a continuous basis.
 - (2) The owner or operator shall calculate and record the additive rate based on a 3-hour block average.
 - (3) If the additive feed rate deviates below the feed rate required (i.e., average additive feed rate observed during the applicable performance test), then the facility shall record the time, date and actions taken to correct the situation and when the additive feed rate is back above the minimum rate required.
- C. The Polishing Scrubber (CE C14) shall maintain an average pressure drop across the wet scrubber that is between 1- and 8-inches water column based on a 12-hour averaging period. The owner or operator shall establish an alarm setting for the purpose of initiating corrective action based on a pressure drop across the wet scrubber of less than 1 inches water column or a pressure drop across the wet scrubber of greater than 8 inches water column. This requirement shall not apply on the days when the equipment the scrubber controls and the scrubber is not in operation.
 - (1) The owner or operator shall record the scrubber pressure drop in inches of water column on a continuous basis.
 - (2) The owner or operator shall calculate and record the average pressure drop across the scrubber based on a 12-hour period. The average pressure drop shall be expressed and recorded as the average of all pressure drop data measured during each 12-hour period.
 - (3) If the pressure drop deviates outside the range specified in Condition 5.C., the owner or operator shall investigate the scrubber controlling emissions and make corrections to the scrubber. The owner or operator shall maintain a record of all corrective actions taken.

- D. The owner or operator shall maintain onsite a copy of the previous performance tests for the scrubbers detailing scrubber pressure drop, scrubber liquid flow rate, and additive feed rate measured during each performance test, which demonstrated compliance with Permit Condition 1.
 - E. The owner or operator shall inspect and maintain the CO₂ Scrubber (CE C07) and Polishing Scrubber (CE C14) according to the manufacturer's specifications. The owner or operator shall keep a log of all maintenance and inspection activities performed on the scrubbers. At a minimum, this log shall include the following:
 - (1) The date that any inspection and/or maintenance was performed on the scrubbers;
 - (2) Any issues identified during inspection and maintenance activities and the date that each issue was resolved; and
 - (3) Identification of the staff member performing the maintenance or inspection.
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6. Continuous Monitoring Systems (CMS)

No continuous monitoring systems are required at this time.

7. Department Review

This permit is issued under the authority of 567 Iowa Administrative Code (IAC) 22.3. The proposed equipment covered by this permit has been evaluated for conformance with the emission limits in this permit; Iowa Code Chapter 455B; 567 IAC Chapters 20 – 35; and 40 Code of Federal Regulations (CFR) Parts 51, 52, 60, 61, and 63 and has the potential to comply. Unless stated elsewhere in this permit, any control equipment covered by this permit shall operate at all times when the emission unit(s) covered by this permit are in operation.

This permit is issued based on information submitted by the applicant. Any misinformation, false statements or misrepresentations by the applicant or by the applicant's representative(s) shall cause this permit to be void.

No review has been undertaken on the engineering aspects of the equipment or control equipment other than the potential of that equipment for reducing air contaminant emissions. The Department assumes no liability, directly or indirectly, for any loss due to damage to persons or property caused by, resulting from, or arising out of the design, installation, maintenance or operation of the proposed equipment.

8. Owner and Operator Responsibility

This permit is for the construction and operation of specific emission unit(s), control equipment, and emission point as described in this permit and in the application for this permit. The permit holder, owner, and operator of the facility shall assure that the installation of the equipment listed in this permit conforms to the design in the application (i.e. type, maximum rated capacity, etc.). No person shall construct, install, reconstruct or alter this emission unit(s), control equipment, or emission point without the required amended permit.

Any owner or operator of the specified emission unit(s), control equipment, or emission point, including any person who becomes an owner or operator subsequent to the date on which this permit is issued, is responsible for assuring that the installation, operation, and maintenance of the equipment listed in this permit is in compliance with the provisions of this permit and all other applicable requirements and that adequate operation and maintenance is provided to ensure that no condition of air pollution is created.

9. Transferability

Unless the equipment is portable, this permit is not transferable from one location to another or from one piece of equipment to another. See Condition 12.A.(2) for notification requirements for relocating portable equipment [567 IAC 22.3(3)"f"].

10. Construction

A. General Requirements:

It is the owner's responsibility to ensure that construction conforms to the final plans and specifications as submitted.

In permit amendments, all provisions of the original permit remain in full force and effect unless they are specifically changed by the permit amendment. If a proposed project is not timely completed, the owner or operator shall seek a permit amendment in order to revert back to the most recent previous version of the permit. The previous, unchanged permit provisions are included in the amendment for your convenience only and are unappealable.

This permit or amendment shall become void if any one of the following conditions occurs:

- (1) The construction or implementation of the proposed project, as it affects the emission point permitted herein, is not initiated within 18 months after the permit issuance date; or
- (2) The construction or implementation of the proposed project, as it affects the emission point permitted herein, is not completed within 36 months after the permit issuance date; or
- (3) The construction or implementation of the proposed project, as it affects the emission point permitted herein, is not completed within a time period specified elsewhere in this permit.

B. Changes to Plans and Specifications:

The owner or operator shall amend this permit or amendment prior to startup of the equipment if:

- (1) Any changes are made to the final plans and specifications submitted for the proposed project; or
- (2) This permit becomes void.

Changes to the final plans and specifications shall include changes to plans and specifications for permitted equipment and control equipment and the specified operation thereof.

C. Amended Permits:

The owner or operator may continue to act under the provisions of the previous permit for the affected emission unit(s) and emission point, together with any previous amendment to the permit, until one of the following conditions occurs:

- (1) The proposed project authorized by this amendment is completed as it affects the emission unit(s) and emission point permitted herein; or
 - (2) This current amendment becomes void.
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11. Excess Emissions

An incident of excess emissions other than as listed in 567 IAC 24.1(1) is a violation and may be subject to criminal penalties according to Iowa Code 455B.146A. If excess emissions are occurring, either the control equipment causing the excess shall be repaired in an expeditious manner, or the process generating the emissions shall be shut down within a reasonable period of time, as specified in 567 IAC 24.1.

An incident of excess emissions shall be orally reported by telephone, electronic mail or in person to the appropriate field office within eight hours of, or at the start of, the first working day following the onset of the incident [See Permit Condition 12.B.(1)]. A written report of an incident of excess emissions shall be submitted as a follow-up to all required initial reports within seven days of the onset of the upset condition [See Permit Condition 12.B.(2)].

12. Notification, Reporting, and Recordkeeping

A. The owner or operator shall furnish the Department the following written notifications:

- (1) In accordance with 567 IAC 22.3(3)"b":
 - a. The date construction, installation, or alteration is initiated postmarked within 30 days following initiation of construction, installation, or alteration.
 - b. The actual date of startup, postmarked within 15 days following the start of operation.
- (2) In accordance with 567 IAC 22.3(3)"f," when portable equipment for which a permit has been issued is to be transferred from one location to another, the Department shall be notified:
 - a. At least 14 days before equipment relocation if the equipment will be located in a nonattainment area for the National Ambient Air Quality Standards (NAAQS) or a maintenance area for the NAAQS.
 - b. At least 7 days before equipment relocation.
- (3) In accordance with 567 IAC 22.3(8), a new owner shall notify the Department of the transfer of equipment ownership within 30 days of the occurrence. The notification shall include the following information:
 - The date of ownership change; the name, address, and telephone number of the responsible official, the contact person, and the owner of the equipment both before and after the ownership change; and the construction permit number(s) of the equipment changing ownership.
- (4) Unless specified, in accordance with a federal regulation, the owner or the owner's authorized agent shall notify the Department in writing not less than thirty 30 days before a required test or performance evaluation of a continuous emission monitor [567 IAC 25.1(7)]. The notification shall include:
 - The time; the place; the name of the person who will conduct the tests; and other information as required by the Department.

If the owner or operator does not provide timely notice to the Department, the Department shall not consider the test results or performance evaluation results to be a valid demonstration of compliance with the applicable rules or permit conditions. Upon written request, the Department may allow a notification period of less than 30 days.

B. The owner or operator shall furnish the Department with the following reports:

- (1) In accordance with 567 IAC 24.1(2), an incident of excess emissions as defined in 567 IAC 20.2 shall be reported within eight hours or at the start of the first working day following the onset of the incident. The report may be made by electronic mail, in person or by telephone.
- (2) In accordance with 567 IAC 24.1(3), a written report of an incident of excess emissions as defined in 567 IAC 20.2 shall be submitted as a follow-up to all required initial reports to the Department within seven days of the onset of the upset condition.
- (3) Operation of this emission unit(s) or control equipment outside of those operating parameters specified in Permit Condition 5 in accordance to the schedule set forth in 567 IAC 24.1.
- (4) In accordance with 567 IAC 25.1(6), the owner or operator of any facility required to install a continuous monitoring system or systems shall provide quarterly reports to the Director, no later than 30 calendar days following the end of the calendar quarter, on forms provided by the Director.
- (5) In accordance with 567 IAC 25.1(7), a written compliance demonstration report for each compliance testing event, whether successful or not, postmarked no later than six weeks after the completion of the test period unless other regulations provide for other notification requirements. In that case, the more stringent reporting requirement shall be met.

- C. All data, records, reports, documentation, construction plans, and calculations required under this permit shall be available at the plant during normal business hours for inspection and copying by federal, state, or local air pollution regulatory agencies and their authorized representatives, for a minimum of two (2) years from the date of recording unless otherwise required by another applicable law (i.e. NSPS, NESHAP, etc.).
- D. Information regarding this permit should be sent to the attention of the following individuals based on the type of information being submitted: change in ownership (Air Quality Bureau Records Center), permit correspondence (Construction Permit Supervisor), stack testing correspondence (Stack Test Coordinator), and reports and notifications (Compliance Unit Supervisor and DNR Field Office). The addresses are:

Air Quality Bureau
Iowa Department of Natural Resources
502 E. 9th St.
Des Moines, IA 50319
Telephone: (515) 725-8200
Fax: (515) 725-9501

DNR Field Office 3
1900 N. Grand Ave., Ste. E17
Spencer, IA 51301
Telephone: (712) 262-4177
Fax: (712) 262-2901

13. Appeal Rights

All conditions within an original permit may be appealed, subject to the appeal rights set forth in 561 IAC Chapter 7. Amended conditions within a permit amendment may be appealed, subject to the appeal rights set forth in 561 IAC Chapter 7. In permit amendments, all provisions of the original permit remain in full force and effect unless they are specifically changed by the permit amendment. The previous, unchanged permit provisions are included in the amendment for your convenience only and are unappealable.

14. Permit History

Table 6 – Permit History

Permit No.	Project No.	Description	Date	Stack Testing
13-A-187	13-041	Combine Permits 06-A-1160-S3 and 06-A-1161-S3 into a Single Permit (Single Emission Point)	04/19/13	Yes
13-A-187-S1	21-149	Add beer column and evaporator to equipment list, convert existing beer column to the side stripper to replace existing equipment.	06/08/21	Yes
13-A-187-S2	22-256	Remove centrifuges (EUs 28-31) from equipment list	11/03/22	Yes

END OF PERMIT